Venkata Lakshmi Parimala Pasupuleti

Senior AI Solutions Engineer · Generative AI · RAG · Agent Workflows · Document Processing · Azure AI · Python · C# · Grounding & Evaluation

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PROFESSIONAL SUMMARY

Senior AI solutions engineer with 8 years shipping production systems and hands-on experience building AI-enabled applications that translate real business needs into working, maintainable solutions. Built **Leasel** - a production AI platform applying **document processing, classification, extraction, knowledge retrieval, and agent workflows** to compliance use cases: OCR ingestion, section-aware chunking, hybrid RAG (TF-IDF + semantic reranking), citation grounding to exact source clauses, and conditional **human review gates** when AI confidence is insufficient - the pattern of grounding + validation + human oversight this role describes. Built **AgentProbe** - an evaluation toolkit that assesses AI solutions for **accuracy, reliability, and consistency**: 5-metric eval engine, LLM-as-judge scoring, CI regression harness, and offline + online evaluation loops. At Entrata (8yr): built and maintained integrations, APIs, and automated workflows across complex enterprise systems - 100% PHI compliance, HIPAA-compliant data handling, zero security incidents. Python, C# (1.5yr), JavaScript/TypeScript, Azure AI, AWS, FastAPI. Active Claude Code user. Case study: github.com/venkatalakshmiparimala/AgentProbe

CORE COMPETENCIES

Generative AI / LLMs | RAG / Knowledge Retrieval | Document Processing & Extraction | Agent Workflows | Classification & Summarization | Prompt Engineering & Structured Outputs | Grounding & Validation | Human Review / HITL | AI Evaluation & Accuracy Assessment | Azure AI Services | Azure OpenAI | Microsoft Power Apps | Python / FastAPI | C# / .NET | JavaScript / TypeScript | APIs / Webhooks / Integrations | Workflow Automation | Application Security & Credential Management | HIPAA / PHI Compliance | Observability & Monitoring | CI/CD & Source Control | AWS (S3 / Lambda / Bedrock) | Docker / Kubernetes | PostgreSQL / SQLite / Redis

EXPERIENCE

Blue Horse Digital

Sep 2026 - Present

AI/ML Engineer (Volunteer) — Virginia, USA

- Building AI-enabled applications and data analysis pipelines; integrating AI components with backend/frontend systems; surfacing business insights into application-facing APIs.

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern

- Built an **agentic workflow system** translating natural-language requests into structured outputs - LangGraph stateful graph with **knowledge retrieval** over FAA corpus, classification, validation guardrails, and conditional routing; deployed MCP server for downstream tool integration.

George Mason University

Aug 2024 - Aug 2025

Research Associate / Graduate Teaching Assistant — Virginia, USA

- Fine-tuned a custom **classification model** (HuggingFace + PyTorch); deployed inference API; evaluated model accuracy and reliability for faculty stakeholders.
- Built multi-source prediction models (Python, R) on 190+ country datasets; applied anomaly detection across 3 research pipelines; delivered technical workshops to 600+ students.

Entrata - PropTech / Real Estate SaaS

Sep 2018 - Aug 2024

Senior Software Engineer — Lehi, UT (Remote)

- **Built and maintained enterprise integrations and automated workflows** across complex multi-tenant systems - HUD income-limit ingestion, document management, and compliance workflows processing 100K+ monthly active users; 80% reduction in manual operational workload via Python/Bash automation.
- **HIPAA-compliant data handling with 100% PHI compliance** - Libsodium-encrypted pipelines, multi-tenant data isolation, access controls at the schema layer; zero security incidents across full AWS S3 migration.
- **C# / .NET experience (1.5yr)** alongside Python/PHP/JavaScript; strong on API design, data modeling, integrations, secure coding, and error handling across application and service boundaries.
- Optimized 200+ SQL queries (30% performance gain); shipped 20+ production modules; mentored 5+ engineers; conducted 300+ peer code reviews; built debugging tools cutting issue resolution time by 40%.

Entrata India - Db Xento Systems

Jan 2017 - Sep 2018

Associate / Trainee Software Engineer — Pune, India

- Built compliance and document management modules (PHP, Python, Java, PostgreSQL); led AWS S3 cloud migration; income limits import (1.6M HUD/Tax Credit records across 5 years).

PROJECTS

Leasel: AI-Enabled Document Processing & Knowledge Retrieval Platform

RAG · Document Processing · Agent Workflows · Grounding · HITL

Production AI platform applying **document processing, extraction, classification, and knowledge retrieval** to compliance workflows: OCR-fallback ingestion, section-aware chunking, **hybrid RAG** (TF-IDF + semantic reranking), **citation grounding** to exact source clauses, missing/conflicting clause detection. **8-agent LangGraph workflow** with conditional routing: Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → **Human Review gate** (fires when AI confidence is insufficient). Core safety rule enforced in code: LLM **extracts and reasons**; human or deterministic logic **makes every regulated decision**. Eval: 93.6% precision, 87.9% F1 on 1,240 test cases; CI regression gate.

Python · LangGraph · FastAPI · TF-IDF · Semantic Reranking · OCR · SQLite · 2026

AgentProbe: AI Solution Evaluation Toolkit

Evaluation · Accuracy · Reliability · Reusable Components

Toolkit for evaluating AI solutions for **accuracy, reliability, consistency, and cost**: **5-metric eval engine** (correctness, groundedness, latency, token efficiency, hallucination rate), LLM-as-judge scoring, CI-integrated regression harness with golden datasets, and **online + offline evaluation loops**. Multi-provider LLM routing (OpenAI/Anthropic/Groq/Ollama); **reusable components and established practices** for responsible AI delivery. Live: github.com/venkatalakshmiparimala/AgentProbe

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) · 2025

AI on FHIR: NLP-Driven Healthcare Data Integration

FHIR · HIPAA · APIs · Clinical NLP · Azure-compatible

Built a Flask + React platform connecting AI models with organizational healthcare systems via **FHIR API integrations** - translating natural-language queries (EN/ES) into structured FHIR calls using LLMs, BERT, and embeddings. **HIPAA-compliant RBAC** with access controls at the data layer; reduced manual clinical retrieval time ~60%. Demonstrates integration of AI capabilities with enterprise systems and regulated data handling.

Flask · React · FHIR APIs · HIPAA RBAC · LLMs · BERT · spaCy · Docker · 2024

EDUCATION

George Mason University **- M.S. Data Analytics Engineering** May 2026

GPA: 3.87 · Virginia, USA · Coursework: ML systems, NLP, cloud infrastructure, statistical modeling, deep learning

Jawaharlal Nehru Technological University **- B.Tech Computer Science & Engineering** 2017